

Part A – Theoretical Foundation (Short Notes & Explanation)

1. What is Statistical Distribution?

A **statistical distribution** is a mathematical function that describes how the values of a random variable are distributed and the probability of each possible outcome. It helps summarize the behavior of data by showing its center, spread, and shape.

Key Characteristics:

- Describes the probability of different outcomes.
- Helps understand data patterns and variability.
- Used for prediction, inference, and decision-making.

Common Statistical Distributions:

- Normal Distribution
- Bernoulli Distribution
- Binomial Distribution
- Poisson Distribution
- Uniform Distribution
- Exponential Distribution
- Log-Normal Distribution
- Power Law Distribution

Applications:

- Business Analytics
- Finance
- Healthcare
- Manufacturing
- Machine Learning
- Risk Analysis

2. What is a Q-Q Plot and Why is it Used?

A **Q-Q (Quantile-Quantile) Plot** is a graphical tool used to compare the distribution of sample data with a theoretical distribution (usually the Normal distribution) or to compare two datasets.

Purpose:

- Check whether data follows a Normal distribution.

- Detect outliers.
- Identify skewness.
- Validate assumptions for statistical tests and regression models.

Interpretation:

- Points close to a straight line → Data follows the theoretical distribution.
- Upward curve → Right-skewed data.
- Downward curve → Left-skewed data.
- Large deviations at the ends → Heavy or light tails.

3. Difference Between Discrete and Continuous Distributions

Feature	Discrete Distribution	Continuous Distribution
Values	Countable	Infinite
Data Type	Integers	Real numbers (decimals)
Probability	$P(X = x) > 0$	$P(X = x) = 0$
Function	Probability Mass Function (PMF)	Probability Density Function (PDF)
Examples	Bernoulli, Binomial, Poisson	Normal, Exponential, Log-Normal

Examples:

- **Discrete:** Number of students, number of defective products.
- **Continuous:** Height, weight, temperature, salary.

4. What is Bernoulli Distribution?

The **Bernoulli Distribution** models a random experiment with only **two possible outcomes**:

- Success (1)
- Failure (0)

If the probability of success is **p**, then:

- $P(\text{Success}) = p$
- $P(\text{Failure}) = 1 - p$

Mean: $\mu = p$

Variance: $\sigma^2 = p(1 - p)$

Example: A coin toss:

- Head = Success
- Tail = Failure

If $P(\text{Head}) = 0.6$, then:

- $P(\text{Head}) = 0.6$
- $P(\text{Tail}) = 0.4$

Applications:

- Pass/Fail testing
 - Spam detection
 - Customer purchase decisions
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5. What is Binomial Distribution?

The **Binomial Distribution** gives the probability of obtaining a fixed number of successes in **n independent Bernoulli trials**.

Conditions:

- Fixed number of trials.
- Two possible outcomes.
- Constant probability of success.
- Independent trials.

Formula:

$$P(X = x) = (nC_x) \times p^x \times (1 - p)^{n-x}$$

where:

- n = Number of trials
- x = Number of successes
- p = Probability of success

Example: A coin is tossed 10 times. Find the probability of getting exactly 6 heads.

Applications:

- Quality control
 - Medical trials
 - Marketing campaigns
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6. Explain Log-Normal Distribution.

A variable follows a **Log-Normal Distribution** if the logarithm of the variable follows a Normal distribution.

If:

$$Y = \ln(X)$$

is normally distributed, then X follows a Log-Normal distribution.

Characteristics:

- Values are always positive.
- Right-skewed.
- Has a long right tail.

Examples:

- Income
- House prices
- Stock prices
- Biological measurements

Applications:

- Finance
 - Insurance
 - Reliability engineering
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7. Explain Power Law Distribution.

A **Power Law Distribution** describes situations where small values occur very frequently while very large values occur rarely but remain significant.

Formula:

$$P(x) \propto x^{-\alpha}$$

where α is a positive constant.

Characteristics:

- Heavy-tailed.
- Scale-free.
- A few observations are extremely large.

Examples:

- Wealth distribution
- Earthquake magnitudes
- Website traffic
- Social media followers

Applications:

- Economics
 - Network analysis
 - Physics
 - Data Science
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8. What is Box-Cox Transform?

The **Box-Cox Transformation** is a mathematical technique used to transform non-normal data into approximately normal data and stabilize variance.

Formula:

For $\lambda \neq 0$:

$$Y = (X^\lambda - 1) / \lambda$$

For $\lambda = 0$:

$$Y = \ln(X)$$

Uses:

- Reduce skewness.
- Improve regression models.
- Stabilize variance.
- Satisfy assumptions of statistical tests.

Applications:

- Machine Learning
 - Linear Regression
 - Forecasting
 - Data Preprocessing
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9. Explain Poisson Distribution with an Example.

The **Poisson Distribution** models the probability of a certain number of events occurring in a fixed interval of time or space when events occur independently at a constant average rate.

Formula:

$$P(X = x) = (e^{-\lambda} \times \lambda^x) / x!$$

where:

- λ = Average number of events
- x = Number of events

Example: A call center receives an average of **5 calls per minute**. The Poisson distribution can be used to calculate the probability of receiving exactly **7 calls** in one minute.

Applications:

- Website visits
 - Customer arrivals
 - Machine failures
 - Road accidents
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10. What is Z-score Probability?

A **Z-score** measures how many standard deviations a data value is from the mean.

Formula:

$$Z = (X - \mu) / \sigma$$

where:

- X = Observation
- μ = Mean
- σ = Standard deviation

Interpretation:

- $Z = 0 \rightarrow$ At the mean.
- Positive $Z \rightarrow$ Above the mean.
- Negative $Z \rightarrow$ Below the mean.
- $\pm 1 \rightarrow$ One standard deviation away.
- $\pm 2 \rightarrow$ Two standard deviations away.

- $\pm 3 \rightarrow$ Three standard deviations away.

Example: Mean = 70

Standard Deviation = 10

Score = 90

$$Z = (90 - 70) / 10 = 2$$

This means the score is **2 standard deviations above the mean**. According to the standard normal distribution, about **97.7%** of observations are below this score.

Applications:

- Standardizing data
- Detecting outliers
- Hypothesis testing
- Quality control

11. Difference Between Probability Density Function (PDF) and Cumulative Distribution Function (CDF)

Feature	PDF	CDF
Full Form	Probability Density Function	Cumulative Distribution Function
Purpose	Shows probability density	Shows cumulative probability
Used For	Continuous distributions	Discrete and continuous distributions
Probability	Area under the curve gives probability	Directly gives cumulative probability
Range	Can exceed 1 (density)	Always between 0 and 1
Graph	Bell-shaped or other density curve	Increasing S-shaped curve

Relationship:

For continuous distributions:

$$F(x) = \int_{-\infty}^x f(t) dt$$

where:

- $f(x)$ = Probability Density Function (PDF)
- $F(x)$ = Cumulative Distribution Function (CDF)

The CDF is obtained by accumulating the area under the PDF curve up to a given value.

Conclusion

Statistical distributions form the foundation of probability and data analysis by describing how data values are spread and how likely different outcomes are. Concepts such as Bernoulli, Binomial, Poisson, Log-Normal, and Power Law distributions help model different types of real-world data, while tools like Q-Q Plots, Box-Cox Transformation, and Z-scores assist in analyzing and preparing data for statistical modeling. Understanding the difference between discrete and continuous distributions, as well as between PDF and CDF, is essential for interpreting probability and making informed decisions in fields such as business analytics, finance, healthcare, engineering, and machine learning.